

Corentin Lunel

Postdoctoral Researcher

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Born on December 24th, 1996
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Research Interests

I am motivated by topics at the interface of mathematics and theoretical computer science. My main interest is computational topology, which is a field in between low-dimensional topology and algorithms. My goal is to investigate the rich relationships between manifolds, knots, surfaces, and graphs from a computational perspective.

Employment and Education

- 2025-** *Charles University, Prague*: Post-doc in collaboration with Martin Tancer.
- 2024-2025** *INRIA UCA*: Post-doc in collaboration with Clément Maria within the Datashape team.
- 2021-2024** *LIGM, Université Gustave Eiffel*: PhD thesis: "Trees, Decompositions, and Knot Theory" under the supervision of Arnaud de Mesmay and Pierre Dehornoy. PHD defended on the 23rd of September 2024.
- Reviewers:
 - David Eppstein, University of California
 - Stephan Tillmann, University of Sydney
 - Examiners:
 - Dominique Attali, Gipsa-lab
 - Livio Liechti, Université de Fribourg
 - Delphine Moussard, Aix-Marseille Université
 - Lionel Pournin, Université Sorbonne Paris Nord
- 2017-2021** *École Normale Supérieure de Lyon*, education at ENSL:
- 2020-2021 ENSL M2 of theoretical computer science.
Internship with Arnaud de Mesmay at *LIGM*: "From decomposing graphs to sweeping knots".
- 2019-2020 ENSL M1 of Mathematics.
Internship with Olga Kravchenko at *Université Lyon 1*: "Le polynôme d'Alexander vu par les graphes bipartis".
- 2018-2019 ENSL M1 of theoretical computer science.
Internship with Uli Wagner at *Institute of Science and Technology Austria*: "Expander graphs and high dimensional Expanders".
- 2017-2018 ENSL L3 of theoretical computer science.
Internship with Arnaud de Mesmay at *Gipsa-lab*: "Réduction monotone de noeuds".
- 2014-2017** *Toulouse*, Higher school preparatory classes at Lycée Pierre de Fermat

Awards

- 2025** Laureate of thesis prize Prix solennel de thèse 2025 from the Chancellerie des Universités de Paris in the general science category.
- 2022** Best PhD student talk at ED MSTIC day.
- 2017** Junior Fermat prize for mathematical research.

Publications

In theoretical computer science, articles are frequently published twice, first in a conference with peer-reviewed proceedings and subsequently in a journal. Even if a paper appears in more than one form, it is only listed once in the following lists. In my field, writers are ordered alphabetically.

Articles in conferences

1. *Hard Diagrams of Split Links*, with Arnaud de Mesmay and Jonathan Spreer, Proceedings of the 41th Symposium on Computational Geometry, 2025. [arXiv](#).
Journal version in Discrete & Computational Geometry (special issue for SoCG 2025), Volume 76, Issue 1, 2026.
2. *Hopf Arborescent Links, Minor Theory, and Decidability of the Genus Defect*, with Pierre Dehornoy and Arnaud de Mesmay, Proceedings of the 40th Symposium on Computational Geometry, 2024. [arXiv](#).
Journal version in Discrete & Computational Geometry (special issue for SoCG 2024), Volume 75, Issue 2, pages 294–319, 2026.
3. *A Structural Approach to Tree Decompositions of Knots and Spatial Graphs*, with Arnaud de Mesmay, Proceedings of the 39th Symposium on Computational Geometry (SoCG 2023), 2023. [arXiv](#).
Journal version in Algebraic & Geometric Topology, Volume 26, Issue 6, pages 2015–2047, 2026.

Articles in journals

4. *Etude d'un invariant des noeuds alternés et mise en oeuvre informatique*, in French, with Hugo Fages and Quentin Rembert, Quadrature 112 (2019) p23-31.

Preprints

5. *Well-quasi-orders on embedded planar graphs*, with Clément Maria, 2025.
arxiv.org/abs/2512.04074.

Presentations

- Géométrie, Topologie et Algorithmes à Grenoble, Grenoble, France, 2026.
- Geometry seminar of Tu Graz, Graz, Austria, 2026.
- Dagstuhl workshop in Intractability in Discrete Geometry and Topology, Dagstuhl, Germany, 2026.
- Séminaire DataShape, Sophia Antipolis, 2026.
- Séminaire MC2, Lyon, France, 12/2025.

- International symposium of Computational Geometry, Kanazawa, Japan, 2025.
- Seminaire Rauzy, Marseille, France, 2025.
- Seminaire MC2, Lyon, France, 01/2025.
- Seminaire ALGCo, Montpellier, France, 2024.
- AATRN Seminar (online), November 2024.
- Geometry & Computing (poster), Marseille, France, 2024.
- International symposium of Computational Geometry, Athens, Greece, 2024.
- Journées du GdR IFM (poster), Grenoble, France, 2024.
- Journées Graphes et Algorithmes, Lyon, France, 2023.
- International symposium of Computational Geometry, Dallas, Texas, USA, 2023.
- SOS Workshop, Dagstuhl, Germany, 2023.
- ED MSTIC PhD Student day, best presentation, Paris, 2022.
- Journée de Géométrie Algorithmique (online), 2022.
- AMS-EMS-SMF Joint Congress of Mathematics, Grenoble, France, 2022.

Teaching

2022-2024	Algorithms course, exercise and practical sessions, 48 hours, ESIPE, first year.
2021-2023	Algorithms and tree data structures, exercise sessions, 22 hours, L2 course at Université Gustave Eiffel.
2021-2022	Lab math-info, exercise and practical sessions, 40 hours, L2 course at Université Gustave Eiffel.

Reviews

- Conference Reviews: SoCG 2023, SoCG 2026, FWCG 2025.
- Journal Reviews: Computing in Geometry and Topology, Journal of Computational Geometry, and Experimental Mathematics.

Skills

Spoken languages

- French, native speaker.
- English, fluent (Certificate in Advanced English, C1).
- German, school level (B1).

Programming languages

- C
- Python
- OCaml
- LaTeX